Amortized Efficiency for Pairing Delegation

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- **New hardness assumption (DBDH)**: Given $(g_1, g_1^a, g_2, g_2^b, g_1^c, T)$ with $g_i \in \mathbb{G}_i$, hard to decide whether $T = e(g_1, g_2)^{abc}$ or random.

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Joux's tripartite Diffie-Hellman: Alice broadcasts $[a]\,P$, Bob broadcasts $[b]\,P$, and Charlie broadcasts $[c]\,P$. Each then computes the shared key

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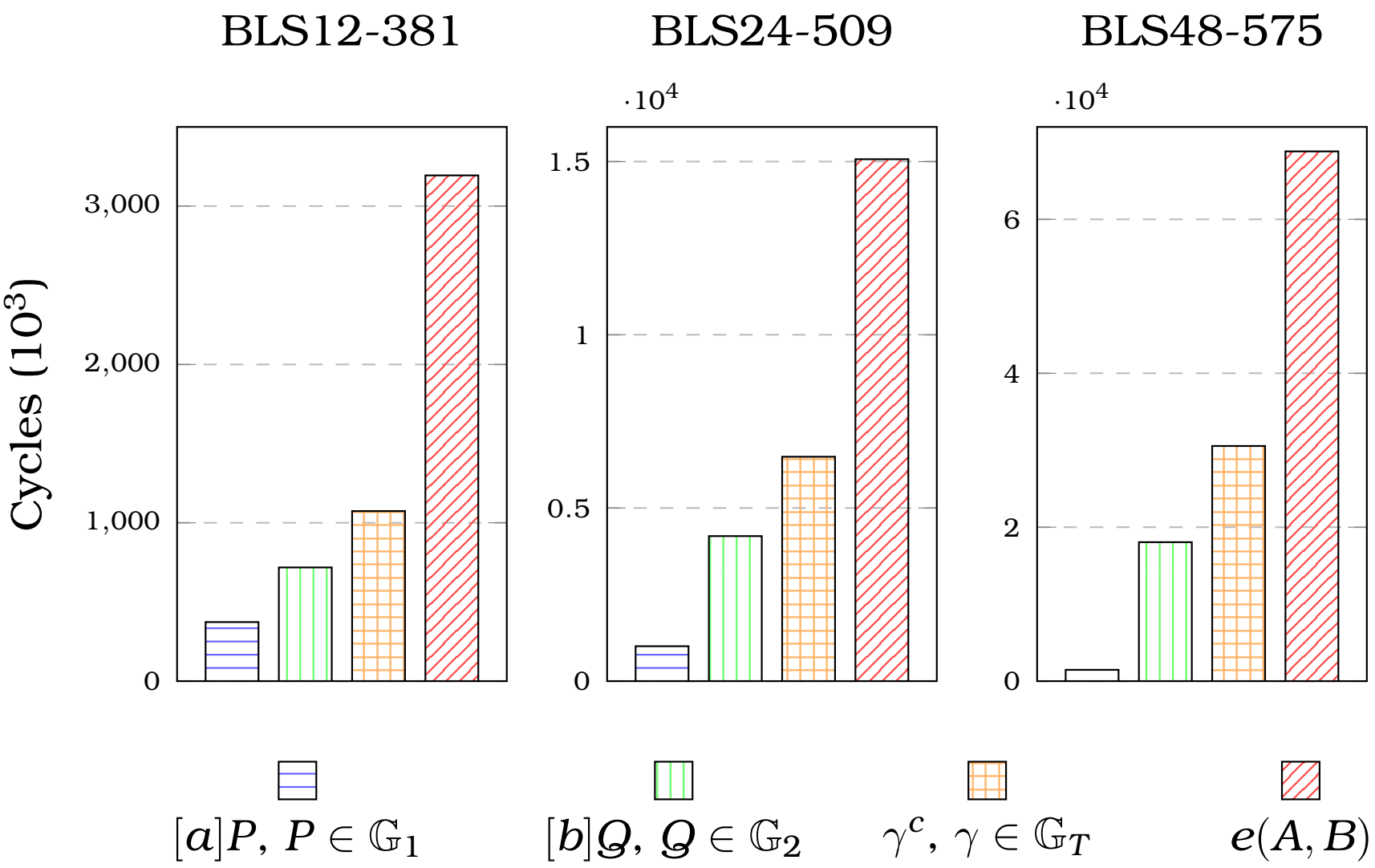
- **zkSNARKs (Groth16)**: Constant-size zkSNARK proofs (3 group elements) with verification reducible to 3 pairings.

Computational Costs

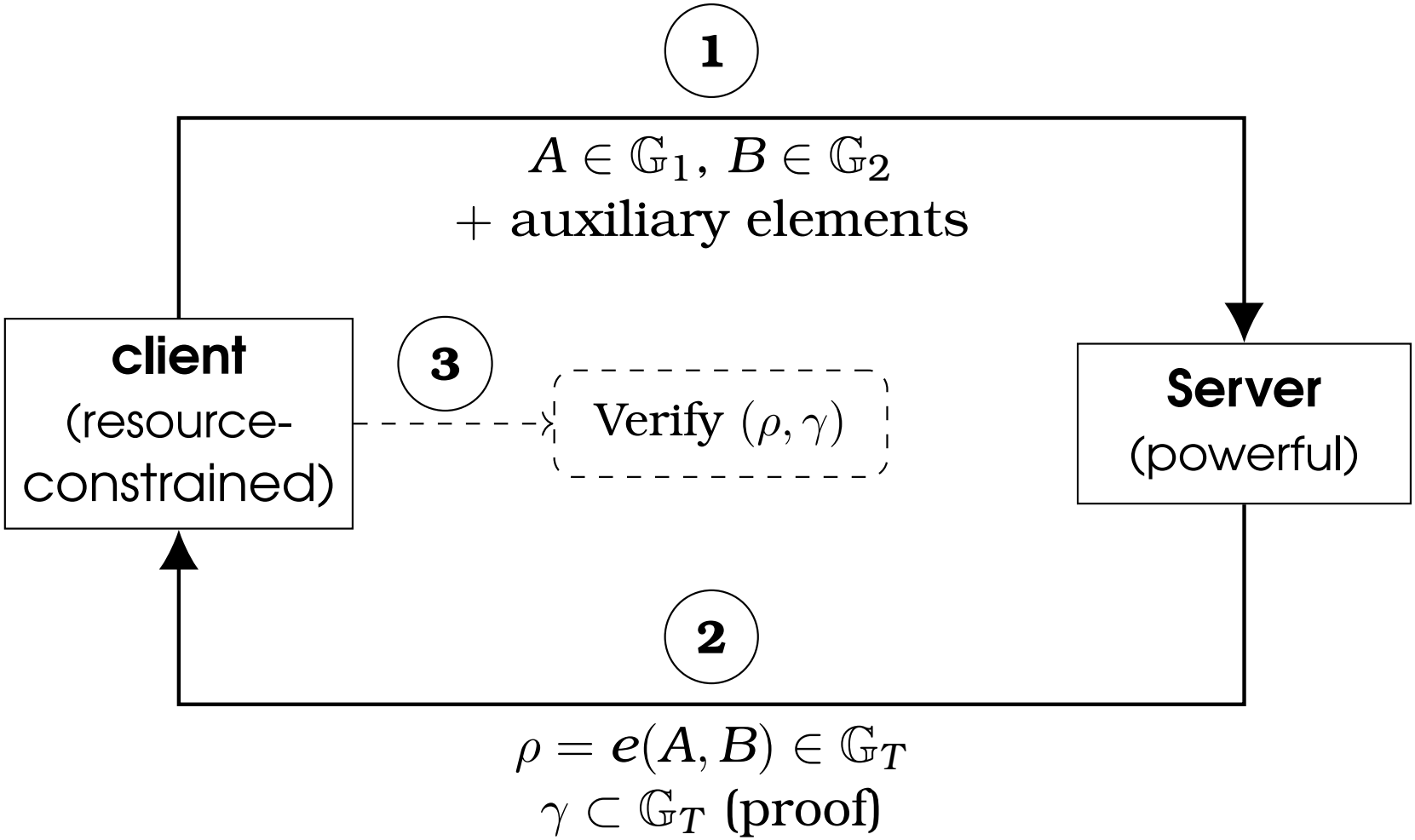


Pairing Delegation

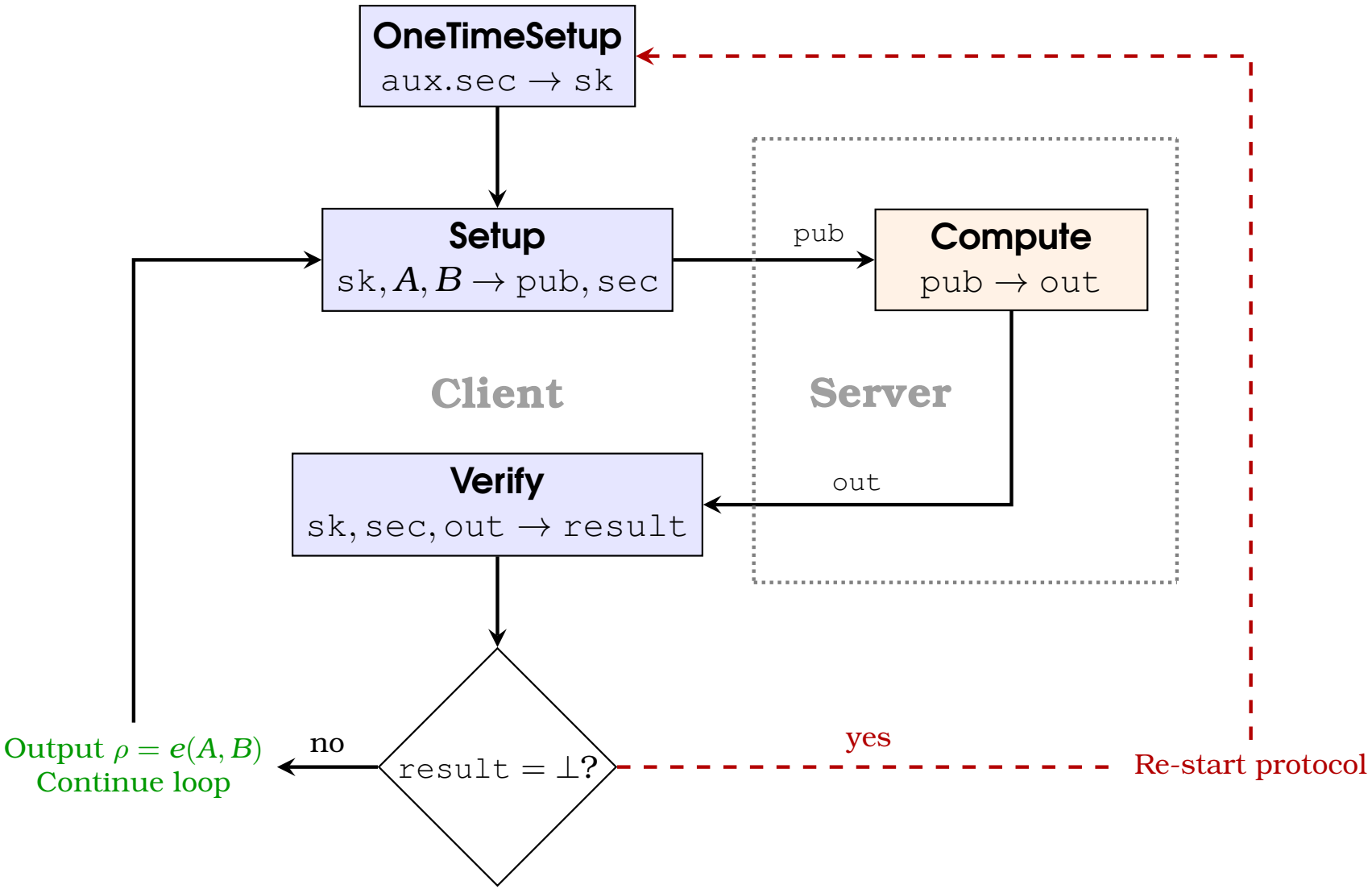
All pairing delegation protocols are

- One-shot:
- Proven unconditionally secure $\rightarrow$ This

One-Shot Pairing Delegation



Sequential Pairing Delegation



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- **Amortized Efficiency**: The client's workload when delegating L pairings must be lower than computing them locally.

$$\frac{\mathsf{cost}(\mathsf{OneTimeSetup}) + L \cdot (\mathsf{cost}(\mathsf{Setup}) + \mathsf{cost}(\mathsf{Verify}))}{L} \leq \mathsf{cost}(e(A,B))$$

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This guarantees unconditional security. Why?

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Lemma 1 (Identifying forgeries)

Let $\mathcal{A}$ be an algebraic adversary, $(\xi, r) \in \mathbb{G}_T \times \mathbb{Z}_q^*$ the client's secret values, $\mathtt{out} = (\rho, \gamma) \in \mathbb{G}_T \times \mathbb{G}_T$ the output of an honest server, and Verify *a verification phase consisting of the checks*

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Proof.

By the closure property of $\mathbb{G}_T$, if $\left(\xi = (\rho^*)^r \cdot (\gamma^*)\right) \wedge (\rho^* \in \mathbb{G}_T)$, then $\mathtt{out}^*$ can be written as a deviation from the honest output: $\rho^* = \rho \cdot \eta$ and $\gamma^* = \gamma \cdot \eta^{-r}$ for any public element $\eta \in \mathbb{G}_T$. Indeed, $(\rho^*)^r \cdot (\gamma^*) = (\rho \cdot \eta)^r \cdot \gamma \cdot \eta^{-r} = \rho^r \cdot \gamma = \xi$. □

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Wait a minute... do we still get unconditional security?

Meet-in-the-middle attack

A toy example:

Let $(\mathbb{G} = \langle P \rangle, +)$ be a cyclic group of prime order q, and (r_1, r_2, u) a tuple of secret values such that $r_1, r_2 \xleftarrow{\$} S \subset \mathbb{Z}_q^*$ and $\varepsilon \xleftarrow{\$} \mathbb{Z}_q^*$. Let $\xi = [\varepsilon]P \in \mathbb{G}$, and consider the public view

$$\mathtt{pub} = \begin{cases} C = [r_1]\xi + X \\ D = [r_2]\xi + Y \end{cases}, \tag{2}$$

where $X, Y \in \mathbb{G}$ are public and $X \neq C, Y \neq D$. $\mathcal{A}$ can attempt to learn the secret tuple from (2) by picking an element

$$\xi^* \in \mathcal{I} = \left\{[t_1^{-1}](C - X) \colon t_1 \in S\right\} \cap \left\{[t_2^{-1}](D - Y) \colon t_2 \in S\right\} \tag{3}$$

and its corresponding indexes r_1^*, r_2^*, which takes up to $2 \cdot |S|$ computations.

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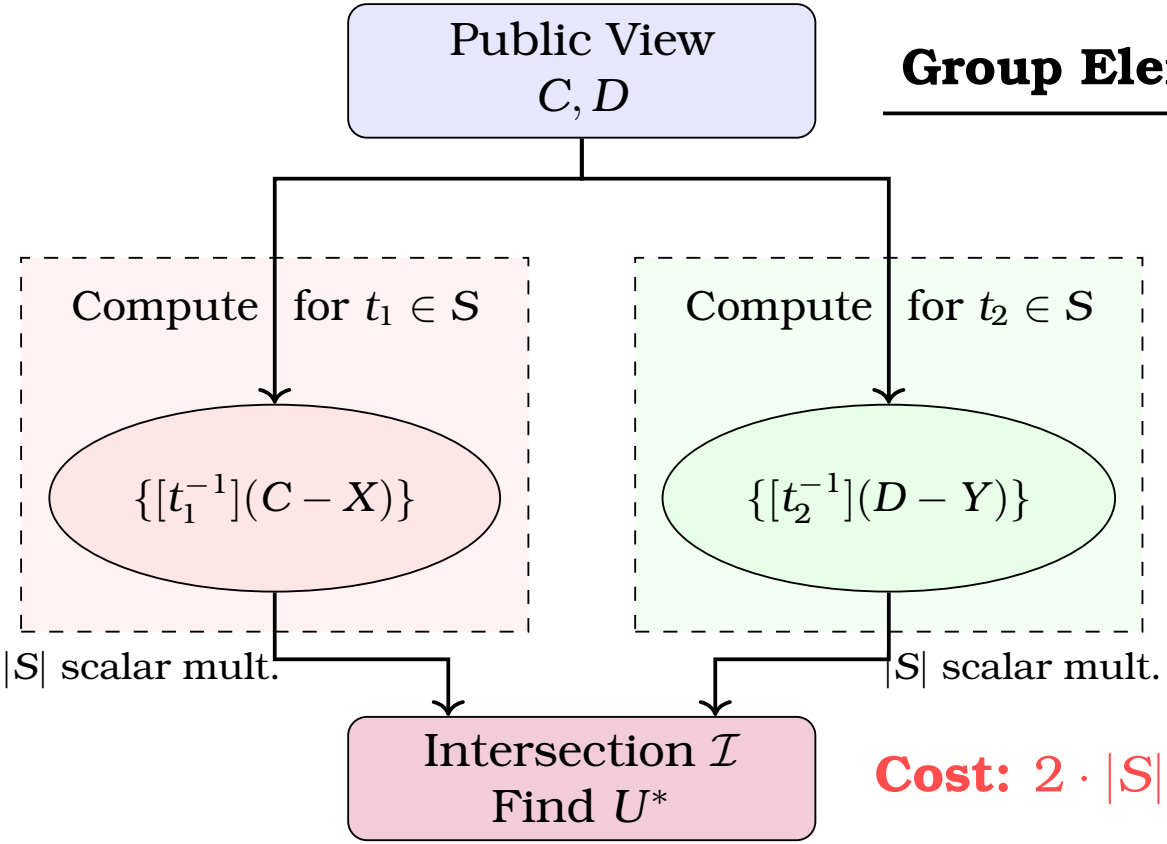

Group Element Intersection Attack

Cost: $2 \cdot |S|$ **scalar multiplications**

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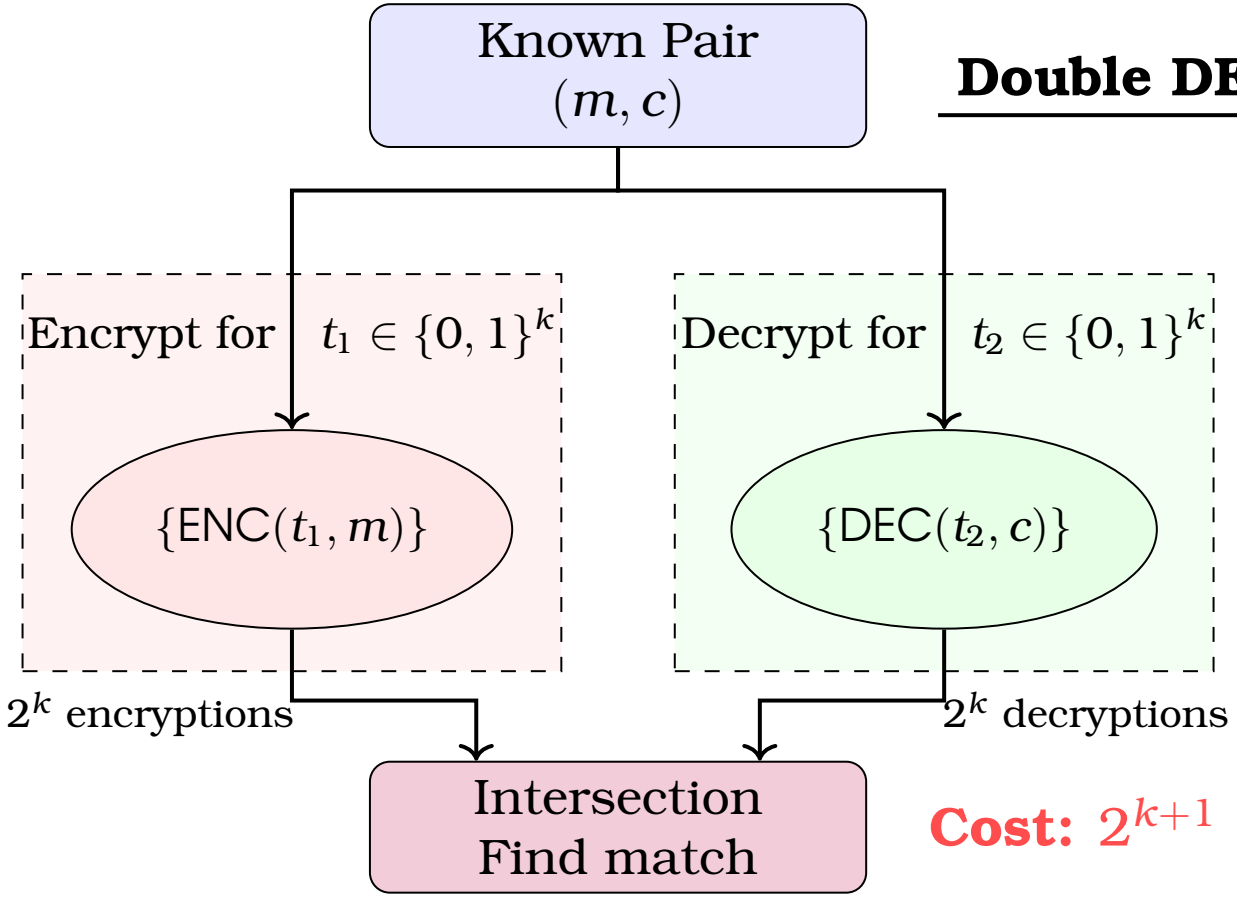

Double DES Meet-in-the-Middle

Cost: 2^{k+1} operations

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- Add a timeout parameter to the protocol.
- Bound $\mathcal{A}$ computationally... but not in the conventional way: Assume $\mathcal{A}$ is not more powerful than the entire Bitcoin network, i.e., $\mathcal{A}$ is allowed to compute 2^κ scalar group multiplications per second (κ matching the current hashrate conversion).

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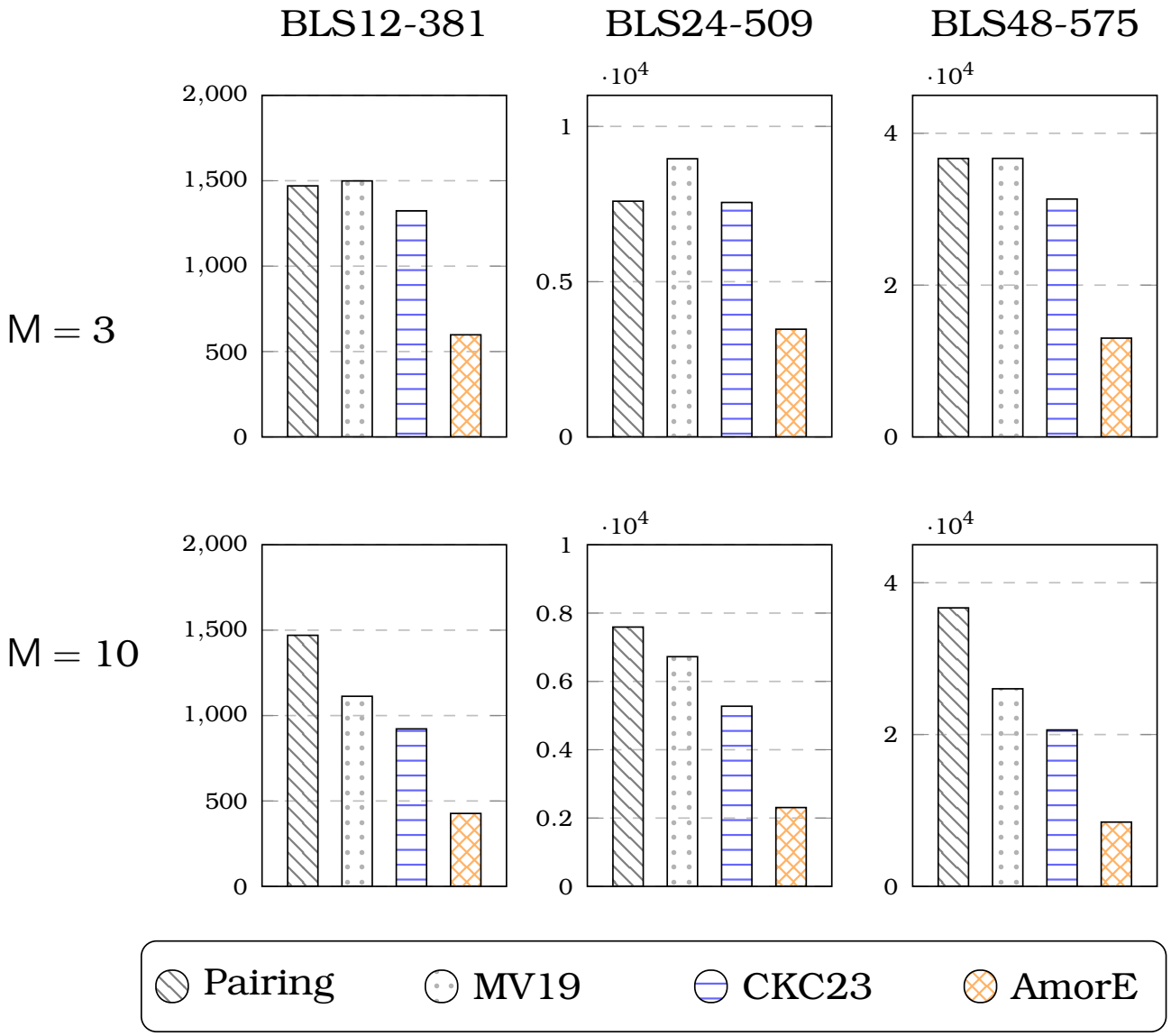
Performance results



Thank you for your attention :)

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